MH1101 Tutorial 6 (Week 7)
Solution

Reference: Sections 3.4, 3.5, 3.6 (Lecture Notes)

1. Evaluate the integral.

(i) $\displaystyle\int_1^2 \frac{x^3+4x^2+x-1}{x^3+x^2}\,dx$

(ii) $\displaystyle\int \frac{4x}{x^3+x^2+x+1}\,dx$

(iii) $\displaystyle\int \frac{x+4}{x^2+2x+5}\,dx$

(iv) $\displaystyle\int \frac{5x^4+7x^2+x+2}{x(x^2+1)^2}\,dx$

(v) $\displaystyle\int \frac{e^{2x}}{e^{2x}+3e^x+2}\,dx$

Solution

(i) The integrand is not proper. By long division,

$$\frac{x^3+4x^2+x-1}{x^3+x^2} = 1 + \frac{3x^2+x-1}{x^2(x+1)}.$$

Write

$$\frac{3x^2+x-1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}.$$

Multiplying both sides by $x^2(x+1)$:

$3x^2+x-1 = Ax(x+1)+B(x+1)+Cx^2 = (A+C)x^2+(A+B)x+B.$

Comparing the coefficients of x^2, x and the constants, we have

$$3 = A+C, 1 = A+B, -1 = B.$$

Thus, $A=2$, $B=-1$, $C=1$, so

$$\frac{3x^2+x-1}{x^2(x+1)} = \frac{2}{x} - \frac{1}{x^2} + \frac{1}{x+1}.$$

So

$$\begin{aligned}
\int_1^2 \frac{x^3+4x^2+x-1}{x^3+x^2}\,dx &= \int_1^2 1+\frac{2}{x}-\frac{1}{x^2}+\frac{1}{x+1}\,dx \\
&= \left[x+2\ln|x|+\frac{1}{x}+\ln|x+1|\right]_1^2 \\
&= (2+2\ln 2+\frac{1}{2}+\ln 3)-(1+2\ln 1+1+\ln 2) \\
&= \frac{1}{2}+\ln 2+\ln 3 \\
&= \frac{1}{2}+\ln 6.
\end{aligned}$$

(ii)

$$\frac{4x}{x^3+x^2+x+1}=\frac{4x}{x^2(x+1)+(x+1)}=\frac{4x}{(x+1)(x^2+1)}=\frac{A}{x+1}+\frac{Bx+C}{x^2+1}.$$

Multiplying both sides by $(x+1)(x^2+1)$:

$4x = A(x^2+1)+(Bx+C)(x+1)=(A+B)x^2+(B+C)x+(A+C)$.

Comparing coefficients,

$$A+B=0, B+C=4, A+C=0.$$

This implies that

$$A=-2, B=2, C=2.$$

Thus,

$$\begin{aligned}
\int \frac{4x}{x^3+x^2+x+1}\,dx &= \int \frac{-2}{x+1}+\frac{2x+2}{x^2+1}\,dx \\
&= -2\ln|x+1|+\int \frac{2x}{x^2+1}\,dx+\int \frac{2}{x^2+1}\,dx \\
&= -2\ln|x+1|+\ln|x^2+1|+2\tan^{-1}x+C.
\end{aligned}$$

(iii) Note that we cannot break x^2+2x+5 down into "smaller" factors (because the quadratic function has no real roots), so it is proper. So the partial fraction decomposition of $\frac{x+4}{x^2+2x+5}$ is just itself. We need to think of a way to integrate this.

Write
$$\begin{aligned}\frac{x+4}{x^2+2x+5} &= \frac{x+1}{x^2+2x+5} + \frac{3}{x^2+2x+5}\\ &= \frac{x+1}{x^2+2x+5} + \frac{3}{(x+1)^2+4}\end{aligned}$$

For the first term, we substitute $u = x^2+2x+5$. This implies that

$$du = (2x+2)\,dx = 2(x+1)\,dx \Longrightarrow (x+1)\,dx = \frac{1}{2}\,du.$$

For the second term, we substitute $v = \frac{1}{2}(x+1)$ which implies that

$$2dv = dx.$$

So

$$\begin{aligned}\int \frac{x+4}{x^2+2x+5}\,dx &= \frac{1}{2}\int \frac{1}{u}\,du + \int \frac{3}{4v^2+4}\cdot 2\,dv\\ &= \frac{1}{2}\int \frac{1}{u}\,du + \frac{3}{2}\int \frac{1}{v^2+1}\,dv\\ &= \frac{1}{2}\ln|u| + \frac{3}{2}\tan^{-1} v + C\\ &= \frac{1}{2}\ln|x^2+2x+5| + \frac{3}{2}\tan^{-1}\left(\frac{x+1}{2}\right) + C.\end{aligned}$$

(iv) Consider the partial fraction decomposition:

$$\frac{5x^4+7x^2+x+2}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}.$$

Multiply both sides by $x(x^2+1)^2$:

$$5x^4+7x^2+x+2 = A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x.$$

$$\Longleftrightarrow 5x^4+7x^2+x+2 = A(x^4+2x^2+1) + (Bx+C)(x^3+x) + Dx^2 + Ex$$

$$\Longleftrightarrow 5x^4+7x^2+x+2 = Ax^4+2Ax^2+A+Bx^4+Bx^2+Cx^3+Cx+Dx^2+Ex$$

$\Longleftrightarrow 5x^4+7x^2+x+2 = (A+B)x^4+Cx^3+(2A+B+D)x^2+(C+E)x+A$

By comparing coefficients:

$$\begin{aligned} A+B &= 5, & (1)\\ C &= 0, & (2)\\ 2A+B+D &= 7, & (3)\\ C+E &= 1, & (4)\\ A &= 2. & (5) \end{aligned}$$

From (10) and (5), we have $B = 3$. From (2) and (4), we have $E = 1$. It now follows from (3) that $2(2) + 3 + D = 7 \Longrightarrow D = 0$. Thus, we have

$$A = 2, B = 3, C = 0, D = 0, E = 1.$$

So

$$\int \frac{5x^4+7x^2+x+2}{x(x^2+1)^2}\,dx = \int \frac{2}{x} + \frac{3x}{x^2+1} + \frac{1}{(x^2+1)^2}\,dx. \quad (6)$$

Clearly,

$$\int \frac{2}{x}\,dx = 2\ln|x| + C_1. \quad (7)$$

Substitute $u = x^2 + 1$. Then $du = 2x\,dx \Longrightarrow x\,dx = \frac{1}{2}\,du$. So

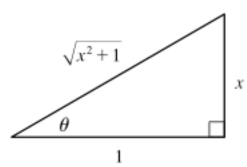
$$\begin{aligned} \int \frac{3x}{x^2+1}\,dx &= \int \frac{3}{u}\cdot\frac{1}{2}\,du \\ &= \frac{3}{2}\int \frac{1}{u}\,du \\ &= \frac{3}{2}\ln|u| + C_2 \\ &= \frac{3}{2}\ln|x^2+1| + C_2. \quad (8) \end{aligned}$$

Let $x = \tan\theta$. Then $dx = \sec^2\theta\,d\theta$, and

$$x^2 + 1 = \tan^2\theta + 1 = \sec^2\theta.$$

So

$$\begin{aligned}
\int \frac{1}{(x^2+1)^2}\,dx &= \int \frac{1}{(\sec^2\theta)^2}\sec^2\theta\,d\theta \\
&= \int \cos^2\theta\,d\theta \\
&= \int \frac{1}{2}(1+\cos 2\theta)\,d\theta \quad \text{(by half-angle formula)} \\
&= \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C_3 \\
&= \frac{\theta}{2} + \frac{2\sin\theta\cos\theta}{4} + C_3 \\
&= \frac{\theta}{2} + \frac{\sin\theta\cos\theta}{2} + C_3 \\
&= \frac{\tan^{-1}x}{2} + \frac{1}{2}\frac{x}{\sqrt{x^2+1}}\frac{1}{\sqrt{x^2+1}} + C_3 \\
&= \frac{\tan^{-1}x}{2} + \frac{x}{2(x^2+1)} + C_3 \qquad (9)
\end{aligned}$$



Substituting (7), (8), (9) into (6):

$$\int \frac{5x^4+7x^2+x+2}{x(x^2+1)^2}\,dx = 2\ln|x| + \frac{3}{2}\ln|x^2+1| + \frac{\tan^{-1}x}{2} + \frac{x}{2(x^2+1)} + C$$

where $C = C_1 + C_2 + C_3$ is some constant.

(v) Let $u = e^x$. Then $du = e^x\,dx \Longrightarrow dx = \frac{1}{e^x}\,du = \frac{1}{u}\,du$. So

$$\int \frac{e^{2x}}{e^{2x}+3e^x+2}\,dx = \int \frac{u^2}{u^2+3u+2}\cdot\frac{1}{u}\,du = \int \frac{u}{(u+1)(u+2)}\,du$$

Let
$$\frac{u}{(u+1)(u+2)} = \frac{A}{u+1} + \frac{B}{u+2}.$$
Multiplying both sides by $(u+1)(u+2)$:
$$u = A(u+2) + B(u+1) = (A+B)u + (2A+B).$$
Comparing coefficients:
$$A+B = 1, 2A+B = 0.$$
Solving these equations:
$$A = -1, B = 2.$$
So
$$\begin{aligned}
\int \frac{e^{2x}}{e^{2x}+3e^x+2}\,dx &= \int \frac{-1}{u+1} + \frac{2}{u+2}\,du \\
&= -\ln|u+1| + 2\ln|u+2| + C \\
&= -\ln(e^x+1) + 2\ln(e^x+2) + C \\
&= \ln\frac{(e^x+2)^2}{e^x+1} + C.
\end{aligned}$$
$\square$

2. Let T_n and M_n be the approximations to the integral $\displaystyle\int_0^1 \cos(x^2)\,dx$ using Trapezoidal Rule and Midpoint Rule respectively.

(a) Find the approximation T_8 and M_8.

(b) Estimate the errors in the approximations of part (a).

(c) How large do we have to choose n so that the approximations T_n and M_n to the integral in part (a) are accurate to within 0.0001?

Solution Let $f(x) = \cos(x^2)$, $a = 0$ and $b = 1$ so that $\triangle x = \frac{b-a}{n} = \frac{1}{n}$.

(a) Since $n = 8$. We have $\triangle x = \frac{1}{8}$,

$$x_0 = 0, x_1 = \frac{1}{8}, x_2 = \frac{2}{8}, x_3 = \frac{3}{8}, x_4 = \frac{4}{8}, x_5 = \frac{5}{8}, x_6 = \frac{6}{8}, x_7 = \frac{7}{8}, x_8 = 1.$$

So

$$\begin{aligned} T_8 &= \frac{\triangle x}{2} \left(f(x_0) + 2(f(x_1) + f(x_2) + \cdots + f(x_7)) + x_8 \right) \\ &= \frac{1}{2 \cdot 8} \left(f(0) + 2(f(1/8) + f(2/8) + \cdots + f(7/8)) + f(1) \right) \\ &\approx 0.902333. \end{aligned}$$

Recall that $\overline{x_i} = \frac{1}{2}(x_{i-1} + x_i)$. So

$$\overline{x_1} = \frac{1}{16}, \overline{x_2} = \frac{3}{16}, \ldots, \overline{x_8} = \frac{15}{16}.$$

Thus,

$$\begin{aligned} M_8 &= \triangle x \left(f(\overline{x_1}) + \cdots + f(\overline{x_n}) \right) \\ &= \frac{1}{8} \left(f(1/16) + f(3/16) + \cdots + f(15/16) \right) \\ &\approx 0.905620. \end{aligned}$$

(b) To apply the error bound, we need to find a constant K such that $|f''(x)| \leq K$ for all $0 \leq x \leq 1$.

Note that

$$f'(x) = -2x \sin(x^2), \quad f''(x) = -4x^2 \cos(x^2) - 2\sin(x^2).$$

So

$$\begin{aligned} |f''(x)| &= |-4x^2 \cos(x^2) - 2\sin(x^2)| \\ &\leq |-4x^2 \cos(x^2)| + |-2\sin(x^2)| \\ &\leq 4 + 2 = 6, \end{aligned}$$

since $|\cos(x^2)| \leq 1$, $|\sin(x^2)| \leq 1$, $|x^2| \leq 1$ for $0 \leq x \leq 1$. Therefore, using $K = 6$, $n = 8$, $a = 0$, $b = 1$ in the error bounds, we have

$$|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{6}{12\cdot 8^2} = 0.0078125,$$
$$|E_M| \le \frac{K(b-a)^3}{24n^2} = \frac{6}{24\cdot 8^2} = 0.00390625.$$

Remark: A better estimate (i.e. smaller upper bound) can be obtained if we have a better (smaller) bound for $|f''(x)|$.

(c) Take $K = 6$ as in part(b)]. Then

$$|E_T| \le \frac{K(b-a)^3}{12n^2} \le 0.0001$$
$$\iff \frac{6(1-0)^3}{12n^2} \le 10^{-4}$$
$$\iff \frac{1}{2n^2} \le 10^{-4}$$
$$\iff 2n^2 \ge 10^4$$
$$\iff n \ge 70.71.$$

So we can take $n = 71$ for T_n.

For E_M, again by taking $K = 6$, we have

$$|E_T| \le \frac{K(b-a)^3}{24n^2} \le 0.0001$$
$$\iff \frac{6(1-0)^3}{24n^2} \le 10^{-4}$$
$$\iff \frac{1}{4n^2} \le 10^{-4}$$
$$\iff 4n^2 \ge 10^4$$
$$\iff n \ge 50.$$

So we can take $n = 50$ for M_n.

3. Let S_n be the approximation of the integral $\displaystyle\int_1^5 \ln x\, dx$ using Simpson's Rule.

(i) Calculate S_8.

(ii) Find a value n such that S_n has error of at most 10^{-6}.

Solution

(i) Let $f(x) = \ln x$, $a = 1$, $b = 5$ so that $\triangle x = \frac{b-a}{n} = \frac{5-1}{8} = 0.5$. We have

$$\begin{aligned}
S_8 &= \frac{\triangle x}{3}\left(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_6) + 4f(x_7) + f(x_8)\right) \\
&= \frac{0.5}{3}\left(f(1) + 4f(1.5) + 2f(2) + 4f(2.5) + \cdots + 2f(4) + 4f(4.5) + f(5)\right) \\
&= \frac{0.5}{3}\left(\ln 1 + 4\ln 1.5 + 2\ln 2 + 4\ln 2.5 + \cdots 2\ln 4 + 4\ln 4.5 + \ln 5\right) \\
&= 4.046655.
\end{aligned}$$

(ii) Let $f(x) = \ln x$. Then $f'(x) = x^{-1} \Longrightarrow f^{(2)}(x) = -x^{-2} \Longrightarrow f^{(3)}(x) = 2x^{-3} \Longrightarrow f^{(4)}(x) = -6x^{-4}$. Since x^{-4} is decreasing on $[1, 5]$,

$$|f^{(4)}(x)| \leq 6(1)^{-4} = 6.$$

Let E_S be the error in using S_n. By the Error bound,

$$|E_S| \leq \frac{K(b-a)^5}{180n^4} = \frac{6(5-1)^5}{180n^4} = \frac{6144}{180n^4}.$$

To ensure the error is at most 10^{-6}, we want to choose n such that

$$\begin{gathered}
\frac{6144}{180n^4} \leq 10^{-6} \\
\Longleftrightarrow n^4 \geq \frac{6144}{180} \times 10^6 \\
\Longleftrightarrow n \geq 76.4354.
\end{gathered}$$

So we can choose $n = 78$, since n must be even for Simpson's Rule.

4. Show that if $f(x) = px^2 + qx + r$ is a quadratic polynomial, then $S_2 = \int_a^b f(x)\,dx$, i.e. Simpson's Rule with $n = 2$ gives the exact value of $\int_a^b f(x)\,dx$.

Solution

Note that $f'(x) = 2px + q$, $f^{(2)}(x) = 2p, f^{(3)}(x) = 0$, $f^{(4)}(x) = 0$. By the Error Bound for Simpson's Rule, using $K = 0$,

$$|E_S| \leq \frac{K(b-a)^5}{180n^4} = 0,$$

so S_2 gives exact value of the integral.

Alternatively, we can also show this from scracth without using Error Bound formula. By Simpson's Rule, we need to show that

$$\begin{aligned} \int_a^b f(x)\,dx &= S_2 = \frac{\triangle x}{3}(f(x_0) + 4f(x_1) + f(x_2)) \\ &= \frac{b-a}{6}(f(a) + 4f((a+b)/2) + f(b)) \end{aligned} \tag{10}$$

Let $g(x) = 1$, $h(x) = x$, $k(x) = x^2$.
Notice that (10) holds for $g(x)$, $h(x)$ and $k(x)$:

- $g(x) = 1$:
$$LHS = \int_a^b g(x)\,dx = \int_a^b 1\,dx = b - a.$$
$$RHS = \frac{b-a}{6}(1 + 4 + 1) = b - a.$$

- $h(x) = x$:
$$LHS = \int_a^b h(x)\,dx = \int_a^b x\,dx = \frac{b^2}{2} - \frac{a^2}{2} = (b-a)(b+a)/2$$
$$RHS = \frac{b-a}{6}(h(a) + 4h((a+b)/2) + h(b)) = \frac{b-a}{6}(a + 4(a+b)/2 + b)$$
$$= (b-a)(b+a)/2.$$

- $k(x) = x^2$

$$LHS = \int_a^b k(x)\,dx = \int_a^b x^2\,dx = \frac{b^3}{3} - \frac{a^3}{3} = (b-a)(a^2+b^2+ab)/3.$$

$$RHS = \frac{b-a}{6}(k(a)+4k((a+b)/2)+k(b)) = \frac{b-a}{6}(a^2+4(a+b)^2/4+b^2)$$
$$= (b-a)(a^2+b^2+ab)/3.$$

Therefore,

$$\begin{aligned}
\int_a^b f(x)\,dx &= \int_a^b px^2+qx+r\,dx \\
&= p\int_a^b x^2\,dx + q\int_a^b x + r\int_a^b 1\,dx \\
&= p\int_a^b k(x)\,dx + q\int_a^b h(x)\,dx + r\int_a^b g(x)\,dx \\
&= p\cdot\frac{b-a}{6}(k(a)+4(k((a+b)/2))+k(b)) \\
&\quad +q\cdot\frac{b-a}{6}(h(a)+4h((a+b)/2))+h(b)) \\
&\quad +r\cdot\frac{b-a}{6}(g(a)+4g((a+b)/2))+g(b)) \\
&= \frac{b-a}{6}(pk(a)+qh(a)+rg(a) \\
&\quad +4(pk((a+b)/2)+qh((a+b)/2)+rg((a+b)/2)) \\
&\quad +pk(b)+qh(b)+rg(b)) \\
&= \frac{b-a}{6}(f(a)+4f((a+b)/2)+f(b))
\end{aligned}$$

$\square$